The TMD2755 provides a complete integrated 3-in-1 sensor solution, reducing board space requirements, and enabling a best-in-class low-profile system design. It combines a low-power

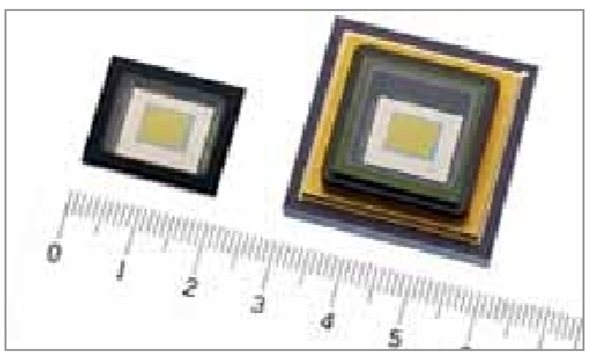
VCSEL emitter (with integrated, factory calibrated driver), an IR photodetector, and an ambient light sensor in a narrow footprint and low profile 0.6mm height package.

By enabling the proximity and light-sensing function in a narrower bezel, the TMD2755 supports increased viewable display area as a proportion of body size, which helps in enhancing the consumer appeal of smartphones in the mid-range segment of the market.

AMS
www.ams.com

Sensor for visible to invisible light

The newly-launched short-wavelength infrared (SWIR) image sensors for industrial equipment have the capability to capture images in both visible and invisible light spectrum in the short-wavelength infrared range.



The sensors employ SenSWIR technology in which photodiodes are formed on an indium gallium arsenide (InGaAs) compound semiconductor layer and are connected via Cu-Cu connection with the silicon (Si) layer, which forms the readout circuit—a design which enables high-sensitivity over a broad range of wavelengths.

This approach enables the SWIR image sensor to deliver seamless image capture ability over a broad range of wavelengths—from visible to invisible light spectrum in the short-wavelength infrared range (wavelength: 0.4µm to 1.7µm).

Sony Corporation
www.sony.net

Sensor for multiple objects

With the introduction of the VL53L3CX, STMicroelectronics has extended the capabilities of its FlightSense Time of Flight (ToF) ranging sensors. The newly released sensor allows measuring distances to multiple objects as well as increasing accuracy.

Unlike conventional IR sensors, the VL53L3CX measures object ranges from 2.5cm to 3m, unaffected by the target colour or reflectance. This allows design-

ers to enable powerful new features into their products, such as occupancy detectors to provide error-free sensing by ignoring unwanted background, foreground objects, or reporting the exact distances to multiple targets within the sensor's field-of-view.

STMicroelectronics
www.st.com

Bluetooth SoC with twin-core processor

Nordic Semiconductor has extended its range of wireless product family with the addition of nRF5340, a Bluetooth LE system-on-chip (SoC) that has not one but two processor cores, making it ideal for professional lighting, advanced wearables, and other complex IoT applications.

There is an increased demand for higher performance of complex wireless applications that has further led to more demand for processing power and memory. A typical single-processor device generally has to decide between higher computational power or efficiency.

The nRF5340 helps in removing this problem since its double core processor has the ability to wake up when it's time to run complex software algorithms, making it battery-friendly. This enables the device to dedicate itself entirely to the application efficiently.

Nordic Semiconductor
www.nordicsemi.com

Frequency flexible crystal oscillators

Here's the Si54x/6x Ultra Series for applications that require low jitter and frequency-flexible clock synthesis. These new oscillators are available with

single, dual, quad, and I2C-programmable frequency options, making them ideal for space-constrained designs that require a combination of different frequencies.

The Si54x/6x series is an ideal solution for 400/600/800G coherent optics and 56G/112G SerDes clocking in optical modules and line cards that require high performance in a small footprint. The new series guarantees ±20 ppm operation over a 20-year operating life.

Silicon Labs
www.siliconlabs.com